

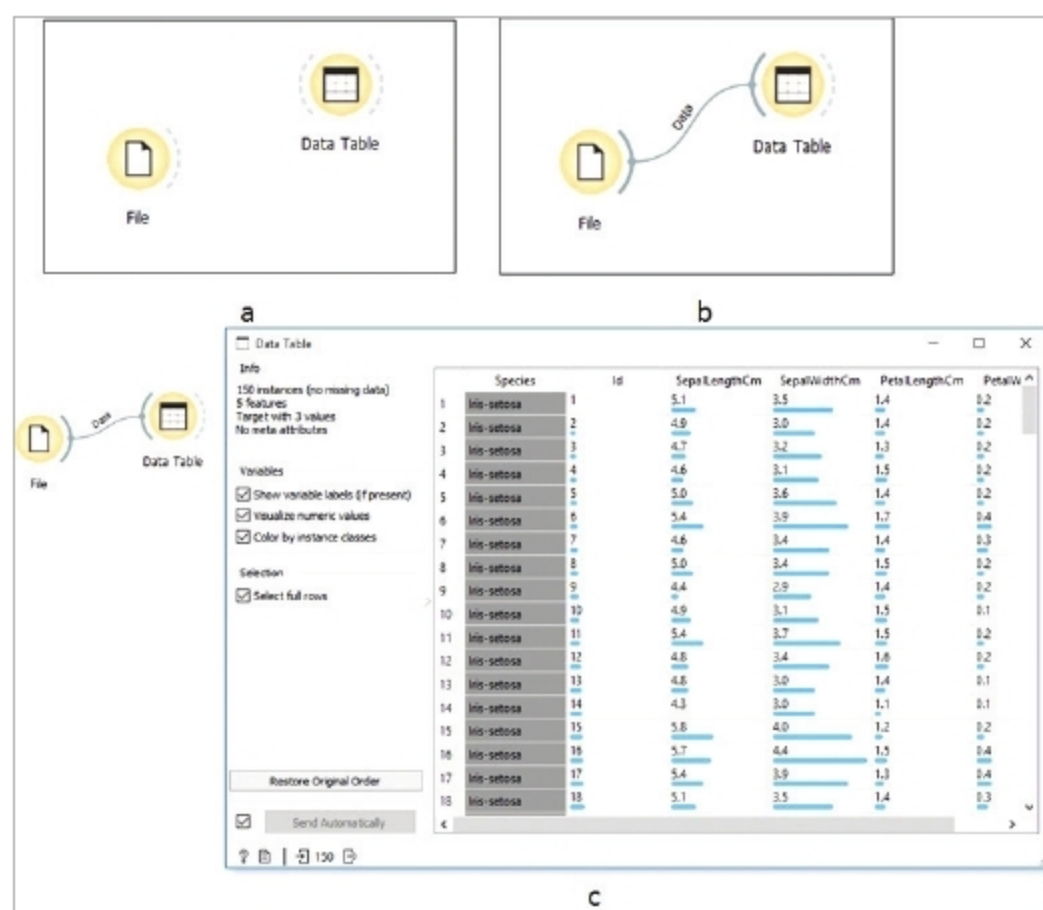


Figure 5: Data exploration spreadsheet



Figure 6: Data set scatter plot

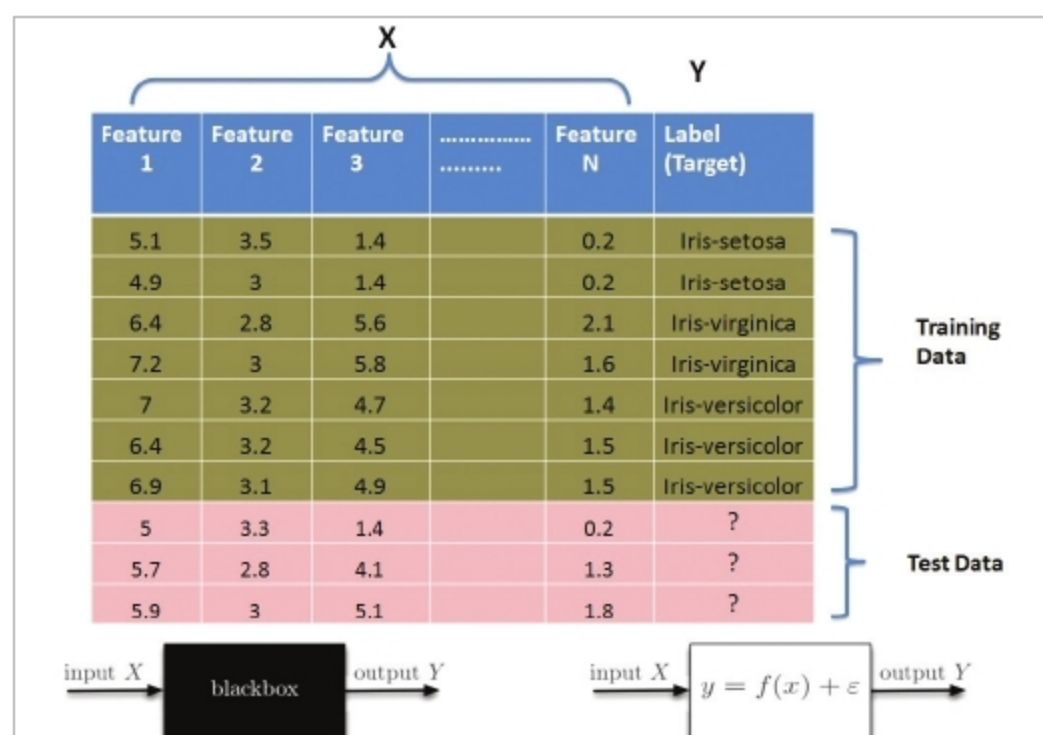


Figure 7: How a machine learning classifier works

sections, namely, categories, widgets and canvas. By default, Orange 3 comes with five categories called Data, Visualize, Model, Evaluate and Unsupervised — each having different widgets, as shown in Figure 2. To perform any kind of task with Orange, we need to construct a workflow by dragging widgets onto the canvas from the respective categories, and connecting them by drawing a line from the transmitting widget to the receiving widget.

Let's build a machine learning classifier using Orange. Follow the steps given below.

Step 1: Download the iris flower data set from <https://www.kaggle.com/ucml/iris>.

Step 2: Load the data set into Orange

Open Orange, drag the *File* widget from the Data category, and place it on the canvas. Data category widgets offer multiple options to load the data into Orange. It is not possible to introduce all those options in this article. So we select the most common widget, i.e., the *File* widget, to load the data as shown in Figure 3a. Now double click the *File* widget to select and load the downloaded data set, as shown in Figure 3b.

The iris data set contains 150 samples with six features. Any feature can be skipped by clicking on the *role* and selecting *skip*. We can make any feature act as a target variable for predictions. Here, we make the feature 'Species' a target.

Step 3: Explore the data set with Orange

Now select and drag the 'Data Table' widget onto the canvas and connect the *File* widget's output to the input of 'Data Table', as shown in Figure 4. Double click the 'Data Table' widget to open the data set in the spreadsheet, as shown in Figure 5. Now to plot the data set, select and drag the 'Scatter Plot' widget from the Visualize category, place it on the canvas and connect it with the *File* widget. Double click on the 'Scatter Plot' widget to get the samples plotted on the graph, as shown in Figure 6.

Step 4: Prepare the training set and testing set

Machine learning classifier takes known samples (training data) and predicts the output for any new unknown sample. At the outset, the machine learning classifier looks like a black box that takes some input sample and predicts the corresponding output label. Mathematically, it approximates the mapping function $f: X \rightarrow Y$ from the input sample feature's values to discrete output labels, as shown in Figure 7.